

Solution Architecture Document

HEALTHCARE ANALYTICS CLOUD FOUNDATION

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DOCUMENT REVISION LOG

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1. Purpose

This Solution Architecture Document defines the high-level architecture for the Enterprise Cloud Foundation for the Healthcare Analytics Cloud Foundation project. It provides the technical blueprint that guides implementation while ensuring alignment with business objectives, security requirements, governance standards, and operational considerations.

2. Business Context

Organization

HealthFirst Diagnostics (fictional) is expanding its analytics capabilities to improve visibility into laboratory operations and support data-driven decision making.

Business Need

The organization requires a secure and governed cloud foundation capable of supporting a future Healthcare Operations Analytics Platform. The platform will enable centralized storage of operational datasets, business intelligence reporting, and future advanced analytics capabilities while maintaining appropriate identity, security, and governance controls.

Business Drivers

- Improve operational visibility through analytics
- Establish a scalable cloud foundation for future analytics initiatives
- Implement consistent governance and security practices
- Minimize operational cost during initial implementation
- Enable future integration with Power BI and Microsoft Fabric

3. Solution Scope

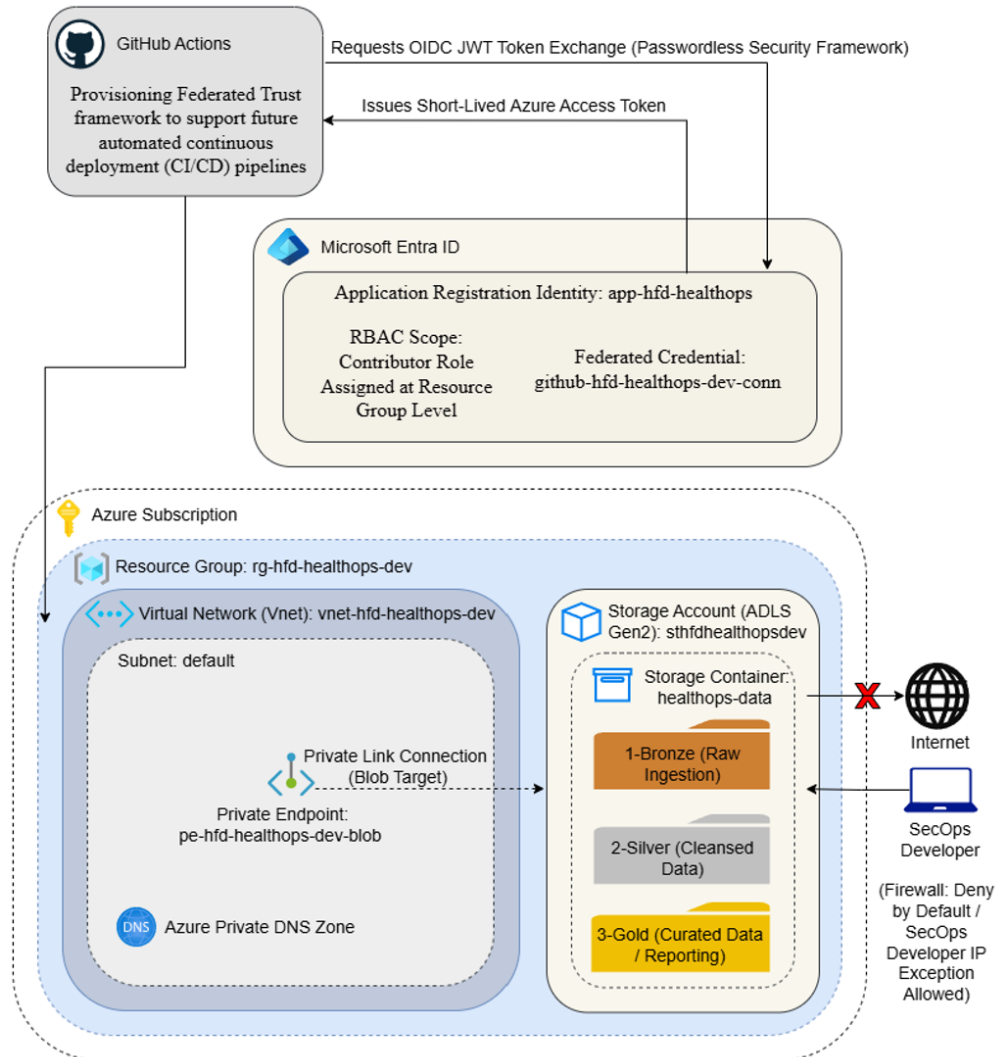
Architecture Scope:

- **Workload Identity Federation (WIF):** Provisioning the passwordless, secretless trust link between GitHub and Azure utilizing OpenID Connect (OIDC)
- **Application Identity Registration:** Creating and configuring the Entra ID App Registration (app-hfd-healthops) to establish a dedicated platform workload security principal
- **Resource Authorization Rules (Azure RBAC):** Establishing localized Role-Based Access Control policies for subscription-level resources to enforce the principle of least privilege
- **Private Cloud Networking & Storage Base:** Establishing the virtual network, custom subnets, private endpoints, private DNS zones, and the ADLS Gen2 Hierarchical Namespace (HNS) data lake
- **Cloud Governance Blueprinting:** Authoring declarative Infrastructure-as-Code (IaC) templates in Microsoft Bicep and organizing metadata tags

Out of Scope:

- **Enterprise Human Identity Management:** Creating or configuring human user accounts, synchronized Active Directory domains, corporate Single Sign-On (SSO), Multi-Factor Authentication (MFA) policies, or Conditional Access rules.
- **Active Downstream SaaS Analytics Engine Implementation:** Constructing live Power BI reports, executing live Microsoft Fabric pipelines, or setting up premium reporting capacities.
- **Production Deployment Orchestration:** Managing active continuous integration and continuous deployment (CI/CD) pipelines, multi-region disaster recovery, cluster scaling, or enterprise system monitoring.

4. High-Level Architecture



5. Compute Paradigm Strategy: Serverless PaaS/SaaS vs. Persistent VMs

A standard development Virtual Machine (VM) incurs un-skippable monthly disk storage and public IP reservation fees even when sitting completely deallocated. By shifting to a serverless blueprint using Azure Data Lake Storage Gen2 (ADLS Gen2), passive storage costs drop to cents per gigabyte, scaling dynamically with active storage consumption. While the companion Microsoft Fabric SaaS compute capacity carries a higher active hourly rate, it consolidates data orchestration, serverless Apache Spark runtimes, and reporting into a single workspace. Because this capacity can be paused to \$0.00/hour during off-hours, it eliminates infrastructure sprawl and results in a lower net Total Cost of Ownership (TCO) for the complete platform ecosystem.

6. Architecture Principles

No.	Principle	Description
1	Serverless Efficiency	Minimize technical debt and management by utilizing native PaaS endpoints rather than active VM operating systems
2	Security by Design	Establish identity, authentication, and authorization before deploying resources
3	Least Privilege	Grant only the permissions required to perform assigned tasks
4	Simplicity	Prefer the simplest architecture that satisfies business requirements
5	Cost Awareness	Evaluate cost implications before provisioning resources
6	Governance	Apply consistent naming, tagging, and documentation standards
7	Network-First Provisioning	Core networking paths and private boundaries must be established prior to provisioning data-bearing assets to prevent public endpoint exposure

7. Design Criteria

No.	Category	Requirement
1	Security	Microsoft Entra authentication
2	Cost	Free Tier where practical
3	Availability	Development environment only
4	Scalability	Support future analytics services
5	Maintainability	Clear documentation
6	Governance	Standard naming and tagging

8. Data Architecture

Resource Name	Performance Tier	Redundancy Level	Primary Service Target
sthfdhealthopsdev	Standard (balanced)	LRS (Locally-Redundant Storage)	Azure Data Lake Storage Gen2 (with /1-bronze, /2-silver, and /3-gold internal directories)

Storage Tiering:

Standard performance combined with Locally-Redundant Storage (LRS) was selected for the development tier to maximize budget efficiency. LRS replicates data three times within a single datacenter zone, matching our development availability SLA while cutting cross-region replication costs by 50%.

Serverless SaaS architecture:

Critically, this storage account is provisioned as an Azure Data Lake Storage Gen2 architecture rather than a flat object storage tier with Medallion structure (Bronze: Raw, Silver: Cleansed, Gold: Curated). This is achieved by explicitly enabling the Hierarchical Namespace (HNS) feature. This design choice establishes a native directory-and-folder hierarchy, allowing analytics processing engines (such as Microsoft Fabric and Power BI) to run high-performance queries and manage data files without the performance degradation associated with traditional flat-file blob systems. This forms the primary ingestion anchor for our analytical data lifecycle.

9. Non-functional Requirements

No.	Category	Requirement
1	Security	Authentication through Microsoft Entra ID
2	Maintainability	Architecture documentation maintained throughout the project
3	Scalability	Support future Power BI and Microsoft Fabric integration
4	Cost	Operate primarily within Azure Free Tier
5	Governance	Resources follow naming and tagging standards
6	Availability	Development environment availability only

10. Design Assumptions

No.	Design Assumption	Value
1	Initial users	10–20 internal users
2	Initial environment	Development only
3	Geographic deployment	Single Azure region
4	Expected workload	Low-volume analytics development
5	Design Assumption	Value
6	Initial users	10–20 internal users

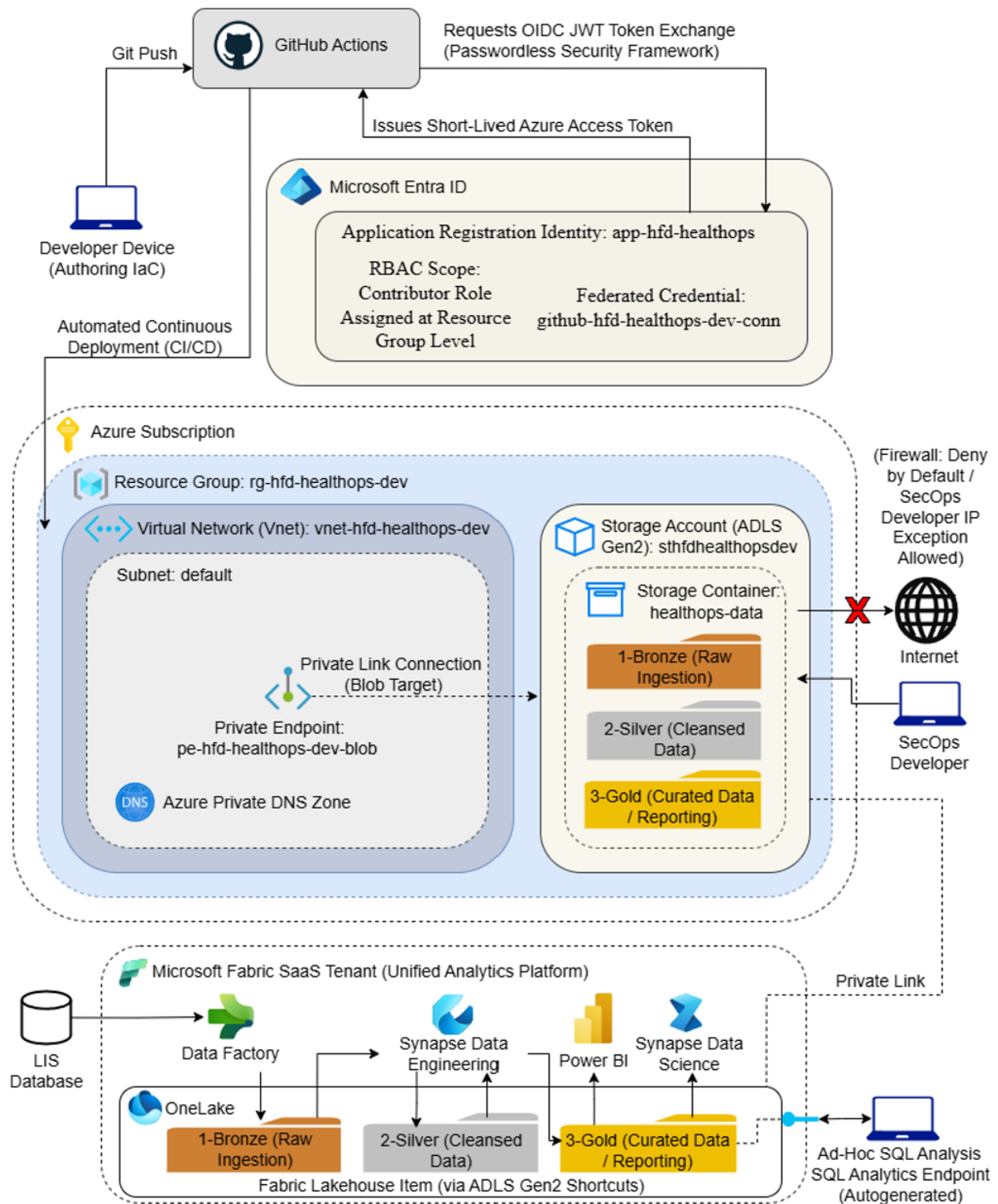
11. Constraints

Refer to Project Charter.

12. Applicable Standards

No.	Standard	Application
1	Standard / Guidance	Application
2	Azure Well-Architected Framework	Overall cloud architecture and design decisions
3	Azure Cloud Adoption Framework	Governance and cloud operating model
4	Microsoft Entra ID Security Best Practices	Identity and access management
5	Azure Naming and Tagging Best Practices	Resource governance
6	Principle of Least Privilege	Security

13. Future Architecture



14. Traceability

No.	Requirement	Solution
1	Secure identity	Microsoft Entra ID
2	Secure storage	Storage Account
3	Governance	Resource Groups
4	Automation	GitHub Actions
5	Cost management	Lifecycle Strategy

15. Technology Selection Rationale

No.	Component	Selected Technology	Alternatives Considered	Rationale
1	Identity	Microsoft Entra ID	Local accounts	Enterprise identity management and RBAC
2	Storage	Azure Storage Account	Azure SQL Database, local files	Appropriate for storing analytics datasets with low operational overhead
3	Compute	Serverless Big-Data Compute (Microsoft Fabric Spark/SQL Runtime)	Azure Virtual Machine, Azure Virtual Desktop, local workstation	Eliminates traditional virtual machine maintenance, security vulnerabilities, and idle hourly costs while providing on-demand analytical processing.
4	Governance	Azure Resource Group	Multiple Resource Groups	Simplifies lifecycle management for a single development environment
5	Automation	GitHub Actions	Azure DevOps Pipelines	Lightweight native workspace provider used to run the validate-azure-identity.yml workflow, confirming Entra ID trust federation without long-lived keys

16. Identity & Access Design

Identity Architecture

Microsoft Entra ID provides identities for users, applications, and automation services. The Healthcare Operations Analytics Platform uses an App Registration to represent the automation identity and a Service Principal (Enterprise Application) to receive Azure RBAC permissions.

Authentication

Authentication uses Microsoft Entra ID. GitHub Actions authenticates to Azure using OpenID Connect (OIDC) federated credentials rather than static client secrets, eliminating long-lived credentials within the repository.

Authorization

Authorization is implemented using Azure Role-Based Access Control (RBAC). **Following the creation of the primary administrative boundary in Sprint 2, the Contributor role is explicitly assigned to the Service Principal at the Resource Group scope.** This enforces the principle of least privilege, ensuring that the automation identity has zero access to other subscription resources or external corporate assets.

Security Principles

- **Least Privilege:** Permissions are scoped tightly at the Resource Group boundary rather than the Subscription level.
- **Passwordless Authentication:** Employs dynamic OpenID Connect (OIDC) token exchanges via Workload Identity Federation (WIF) to eliminate exposed keys.
- **Centralized Identity Management:** Governed entirely via enterprise Microsoft Entra ID policies.
- **Role-Based Authorization:** Restricts resource configurations to native Azure built-in standard roles.

Architectural Cross-Reference

The end-to-end cryptographic OIDC handshake sequence, dynamic Workload Identity Federation (WIF) token flow, and scoped Azure RBAC boundary detailed above are visually mapped under Section 4 (High-level Architecture).

For a look at how this trusted identity orchestrates downstream data processing, see Section 13 (Future Architecture).